

Narrative CV

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How to use this document

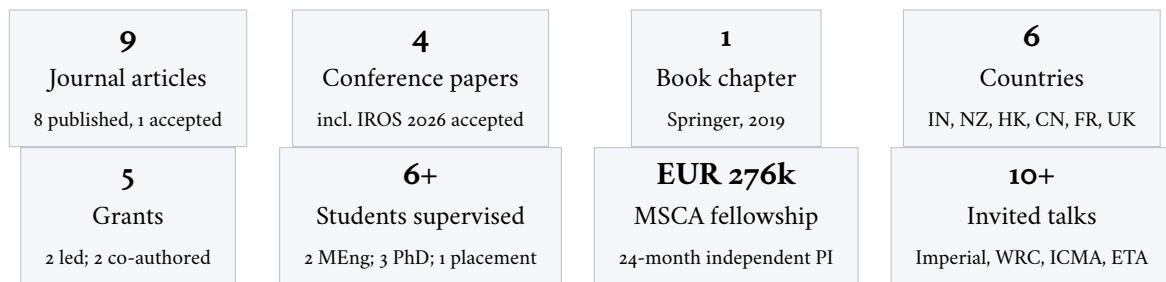
This file contains **two narrative CV versions** plus **visual summaries**. Narrative CVs are prose-based accounts of achievement—not lists—used by UKRI, Horizon Europe, fellowships, and promotion panels.

Version	What it is	When to use	Typical limit
Short (see p. 4)	Concise overview: identity, CASREx, career arc, impact in ~370 words.	Form word boxes; MSCA summaries; cover letters; reviewer first pass; LinkedIn / website bio.	300–400 words
Long (see p. 3)	Full narrative with quantified evidence, contextualised outputs, collaboration, teaching, and future direction.	UKRI narrative CV modules; fellowship applications; promotion dossiers; European Talent Academy; interview preparation.	400–800 words per module (check funder)
Figures (pp. 2–3)	Metrics dashboard, career timeline, research-domain map.	Paste into PDF applications; present in seminars; use <code>narrative-cv.html</code> for interactive viewing.	Visual supplement

Tip from narrative-CV training (Ines)

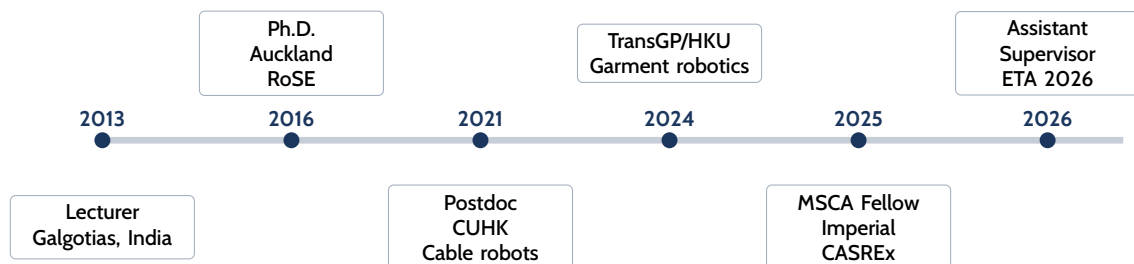
Narrative CVs benefit from **visualisation**: a metrics snapshot and career map help reviewers grasp breadth quickly before reading prose. Use the figures on the following pages alongside either the short or long text.

Research impact at a glance

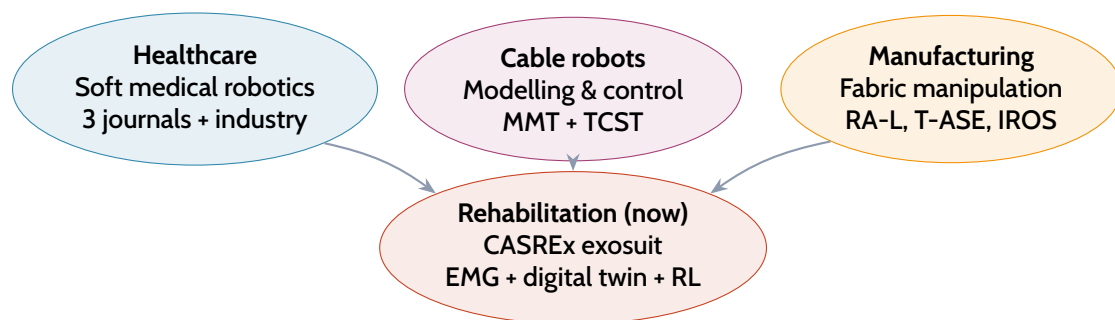


Quantified from own-bib.bib, cv-simple.pdf, and Profile.pdf (March 2026). Journal count: 8 published + 1 accepted (IEEE T-ASE). Conference count: 3 published + 1 accepted (IROS 2026). Grants: MSCA (EUR 276,187.92, lead), BvW (NZD 39,000, lead), 2 ITF co-authored, 1 travel grant.

Career timeline



Research domains and how they connect



Long narrative CV

Research identity and expertise

My distinctive contribution sits at the crossover of **soft and cable-driven robotics**, **human-centred control**, and **learning-enabled systems** for healthcare, rehabilitation, and manufacturing. Across **9 journal articles** (8 published, 1 accepted), **4 conference papers**, and **1 Springer book chapter**, I have built a coherent line of work: model deformable systems, estimate state under uncertainty, and close the loop with adaptive control and learning.

I am Marie Skłodowska-Curie Postdoctoral Fellow at Imperial College London, where I lead **CASREx**—a 24-month, EUR 276,187.92 Horizon Europe fellowship (Grant No. 101205678) and the most independent stage of my career.

Current fellowship

After stroke, many people can reach but cannot keep the arm stable when bumped or loaded. The UK RATULS trial (n=770) showed conventional robot therapy did not consistently beat usual care. CASREx develops a soft, cable-driven upper-arm exosuit that targets this *stability* problem—not only movement—using human experiments, muscle recordings (EMG), a digital twin, and machine learning. The project extends my prior soft-robotics and control work into rehabilitation with direct societal relevance.

At Imperial's Morph Lab, I also supervise **2 MEng placement students**, teach robotics simulation (DESE71002), deliver seminars, run Morph Lab training, and pursue **2 Imperial Connect Fund** collaborations. I was selected for the **European Talent Academy 2026** (Imperial, TU Munich, Politecnico di Milano).

Career development with contextualised outputs

Ph.D., Auckland (2016–2021): Built RoSE, a robotic soft esophagus for stent testing; **3 journal articles**, **2 conference papers**, **1 book chapter**. Industry contract with BvW Holding (NZD 39,000) led to *Soft Robotics* (2021) and *IEEE TIE* (2021)—my first translation from lab to external partner.

Postdoc, CUHK (2021–2024): Cable-robot modelling and control; first-author *Mechanism and Machine Theory* (2025) and *IEEE TCST* (2023); co-authored **2 ITF grant proposals**; mentored UG, placement, and PhD students; visiting fellowship at LS2N, Nantes.

TransGP / HKU (2024–2025): Led garment-robotics programme; built industrial fabric testbeds; outputs include *IEEE RA-L* (2025), *IEEE T-ASE* (2026, accepted), *IEEE IROS* (2026, accepted), and *IEEE ICMA* (2025)—demonstrating team leadership, hardware delivery, and publication in parallel.

Doctoral contribution: RoSE (*Soft Robotics*, 2021)

Endoprosthetic stent migration and swallow dysfunction are difficult to study reproducibly in patients. During my Ph.D. at Auckland, I co-developed **RoSE**, a pneumatic soft-robotic esophagus for controlled in vitro testing of covered self-expandable metal stents. As first author, I contributed to the layered actuator design and fabrication workflow, programmed symmetric and peristaltic actuation protocols, and experimentally validated stent radial force, migration displacement, and intrabolus pressure signatures under manometry—including industry-funded testing (BvW Holding, NZD 39,000). Published in *Soft Robotics* (2021), RoSE established an early benchtop approach to physiologically structured stent evaluation. Its continuing relevance is reflected in subsequent biomimetic esophageal motility research (*Nature Communications*, 2026), which cites RoSE and references its circumferential pneumatic design while advancing independent longitudinal–circumferential actuation and biohybrid tissue modelling.

Impact, collaboration, and community

Impact: Medical-device testing (BvW); garment-manufacturing automation (TransGP); upper-limb rehabilitation (CASREx). **Collaboration**: **6 countries** (India, New Zealand, Hong Kong, China, France, UK); industry and academic partners; MSCA and ETA networks. **Community**: Reviewer for *IEEE RA-L*, *ICRA*, *ASME JMR*; **10+** invited talks including World Robotics Conference 2025 and IEEE ICMA 2025 workshop; supervision of **6+** students (2 MEng, 3 PhD at HKU, 1 placement at CUHK).

Recognition and future promise

MSCA fellowship; European Talent Academy 2026; Best Conference Paper, RiTA 2017; GATE 99.12% (2011); Marie Curie Alumni Association. Near-term: validate CASREx as an impedance-aware rehabilitation framework. Medium-

term: establish a research programme on human-centred soft and cable-driven robots across healthcare and manufacturing.

Short narrative CV (370 words)

I am a robotics researcher developing intelligent machines for healthcare, rehabilitation, and manufacturing—with 9 journal articles, 4 conference papers, and research across 6 countries. My work addresses settings where the body or handled material is soft, deformable, and easily disturbed.

I lead **CASREx**, a Marie Skłodowska-Curie Horizon Europe fellowship at Imperial College London (EUR 276,187.92; 24 months; Grant No. 101205678) on a cable-driven soft exosuit for upper-arm muscle-impairment rehabilitation. Everyday stability—carrying a cup without spilling when bumped—depends on how the brain adjusts arm stiffness. After stroke, this is often impaired. CASREx combines human experiments, muscle-signal recordings, a digital twin, and machine learning to translate healthy stiffness strategies into adaptive patient assistance.

My career has moved deliberately: Ph.D. at Auckland (RoSE soft esophagus; NZD 39,000 industry funding; *Soft Robotics* and *IEEE TIE*); postdoc at CUHK (cable-robot modelling; *MMT* 2025, *IEEE TCST* 2023); Senior Research Engineer at TransGP (led fabric-robotics programme; *IEEE RA-L* 2025, *T-ASE* 2026 accepted, *IROS* 2026 accepted). I now supervise 2 MEng students at Imperial, mentor doctoral researchers at HKU, review for major robotics venues, and was selected for the European Talent Academy 2026.

My aim is a long-term programme on human-centred soft robotics—helping people recover stability, and building automation that works with the real world.

Sources: `cv-simple.pdf`, `Profile.pdf`, `own-bib.bib`. Interactive figures: [narrative-cv.html](#)